

Introduction to Figurate Numbers

An Introduction by Joshua Aaron

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1 Don't be Such a Square!

I remember in grade school learning the concept of “squaring” a number, and even “cubing,” switching between dimensions and unites with just a movement of a number into the upper right. Squares and cubes are great and all, but I remember being curious as to why we never talked about other types of shapes? Surely triangular numbers are just as interesting? What about tetrahedral (3-D triangles) or pyramidal numbers? Here we are going to begin an adventure into the world of *figurate numbers* and their pertinent *gnomons*!¹ Though unfortunately Gnomon(s) will have to wait until next time, figurate numbers, or numbers that represent shapes, will serve as useful tools by which we can look at composition of functions and consider functions as mappings between infinite sets. We haven't really dealt with these issues when dealing with ciphers and modifying alphabets. We have already seen a little function composition, and we can certainly think about the infinite set of words, but much of what we were dealing with was going between alphabets—here we will be going back to our old friends, the numbers. If you want to keep the coding idea alive in your mind a bit longer, it might be helpful to realize that every time we write a number we are really encoding the relationship between a value (or in the case of a multi-digit number, the sum of values) and a symbol (or a list of symbols for a multi-digit number).

2 Fun with Dots

As the wordy introduction said, lets have another look at the square numbers, shall we? We know that we can depict or represent these numbers as squares as follows:

$$\text{square}(2) = 2^2 = 4 = \begin{array}{cc} \cdot & \cdot \\ \cdot & \cdot \end{array} \quad \text{and} \quad \text{square}(3) = 3^2 = 9 = \begin{array}{ccc} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{array}$$

Indeed this may have been how square numbers were taught to you initially! In particular I was taught multiplication (of whole numbers) using rectangles on giant peg boards (Montessori education can be great for visualizing math!), so with giant pegboards in mind, how might triangles and triangular numbers look? Maybe something like:

$$\text{triangle}(2) = 1 + 2 = 3 = \begin{array}{cc} \cdot & \\ \cdot & \cdot \end{array} \quad \text{and} \quad \text{triangle}(3) = 1 + 2 + 3 = 6 = \begin{array}{ccc} & \cdot & \\ \cdot & \cdot & \\ \cdot & \cdot & \cdot \end{array}$$

Okay enough words and warming up, lets get into the interesting questions!

¹After an extensive search involving your teachers climbing through abandoned catacombs in search of ancient Greek mathematical and astrological texts, it appears that the world (and history) has not decided how to pluralize the word gnomon. This appears to be quite common when a word is very infrequently used and of Greek/Latin origin. Gnomon as a plural is probably also acceptable to the spelling gods, but it might be confusing here so we will stick with the normal s-pluralization.

1. Can you come up with a more general expression for $triangle(k)$ where k is a natural number? ² As a hint, maybe it would help to think of each triangle as a sum from 1 to k ...
2. or maybe there is a recurrence that can be used? Put another way, can we write $triangle(k)$ in terms of $triangle(k-1)$ or perhaps other previous triangular numbers? (there are lots of cool ways of doing this...)
3. Can you write $triangle(4k)$ in terms of $triangle(k)$? Is there an associated picture that makes the relationship make sense? What does it look like? How about in terms of $triangle(2k)$? Any other schmancy³ ideas?

3 N-gon Numbers

Now that we have the idea of square and triangular numbers, it only makes sense to ask about pentagonal and hexagonal numbers right? The first thing to notice is that every regular n -gon (polygon with n sides, where n is a natural number) can be split into n triangles. Pause and ponder for a minute—maybe draw some shapes to convince yourself! With this in mind, see how the following questions feel:

1. Can you come up with a function that uses triangular numbers to calculate pentagonal numbers? Can you use that to come up with a general function for the k th pentagonal number in terms of only k ? (I would call this number $penta(k)$ or something like that, just so its clear what you are talking about.)
2. Can you come up with a similar definition for Hexagonal Numbers?
3. Are there any hexagonal numbers that are also triangular numbers? Which ones? Are there any square hexagonal numbers?
4. Now for the super fulfilling question: can you come up with a function $ngon(n, k)$ which for natural numbers n and k gives us the k th n -gon number? So if $n = 3$ and $k = 4$ it would give us the 4th triangular number (that is, $ngon(3, 4) = 10$ depending on how you start counting).
5. For all of the above, what does the 0th or 1st n -gon number look like? (Is it 0? or 1? or n ? something else entirely?) This is your definition so it is up to you—but definitely justify why you chose that! ⁴
6. Using your newfound $ngon(\cdot, \cdot)$ function can you find a condition for n such that n is a square, triangular, and pentagonal number? (If you can do it without that function that's fine too!)
7. Given a number n can you generate a function $leastsides(k)$ that returns the type of shape with the fewest amount of sides that can be used to depict that number? How about $mostsides(k)$ which would return the shape with the most sides you can make with n points?

²“where k is a natural number” can also be represented symbolically as $k \in \mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$

³a technical term of great semantic importance

⁴Taste/convenience are acceptable answers only if you have run out of other reasons...